

Muddiest points: Feb 17

- For viscoelasticity how are stress and strains equal to each other for two separate systems?

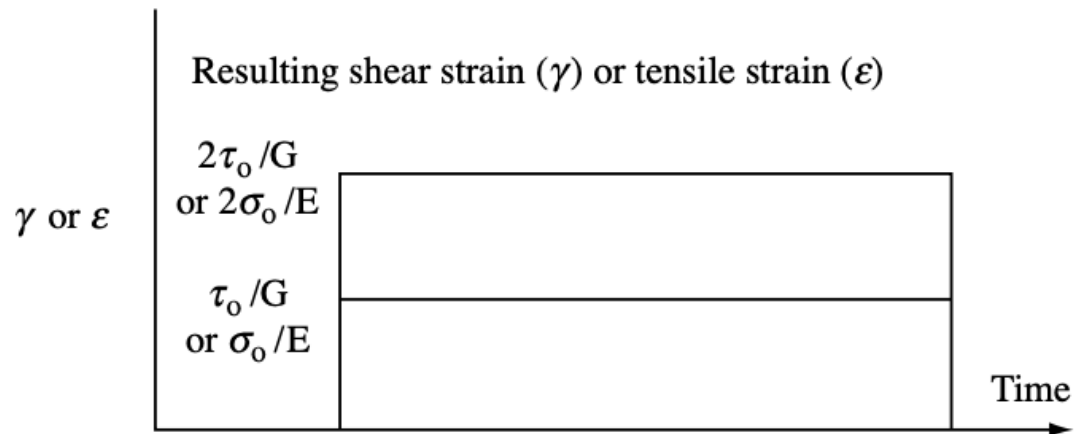
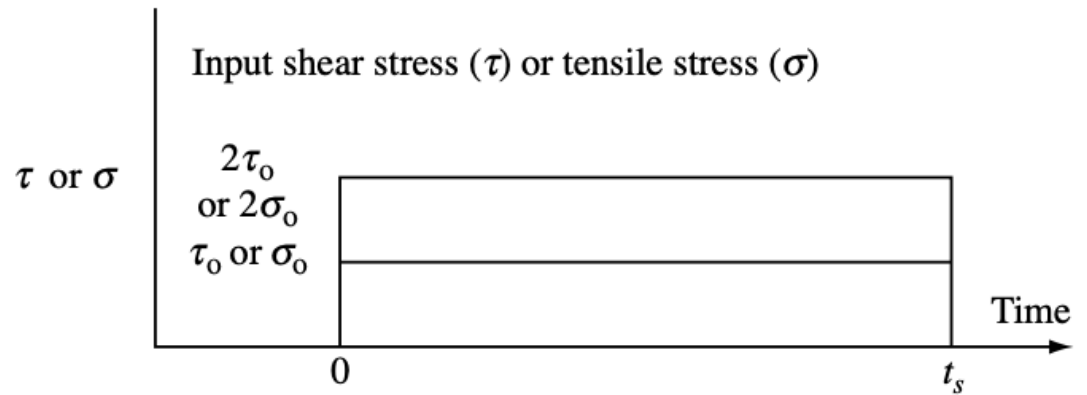
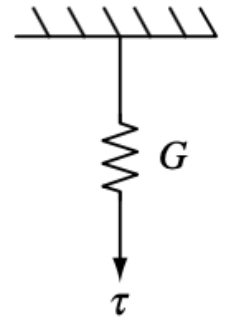
Muddiest points: Feb 17

- What do the variables in the following equations correspond to?

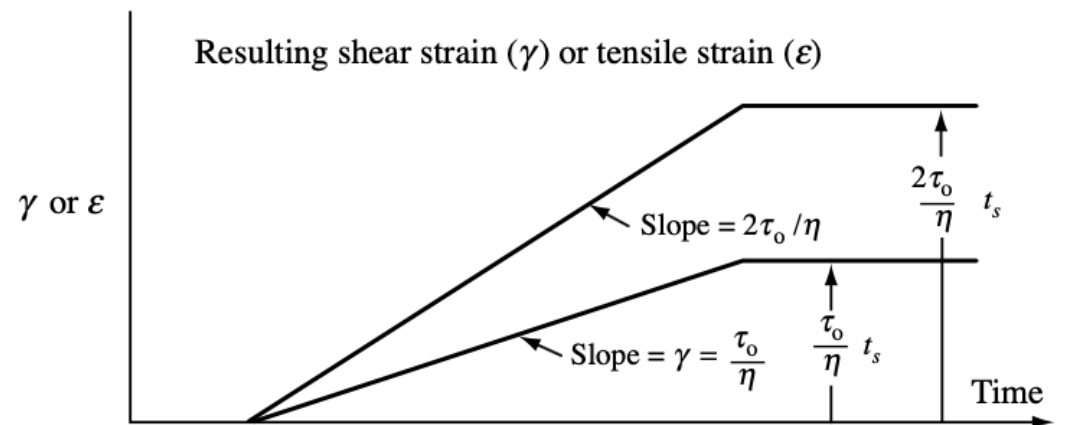
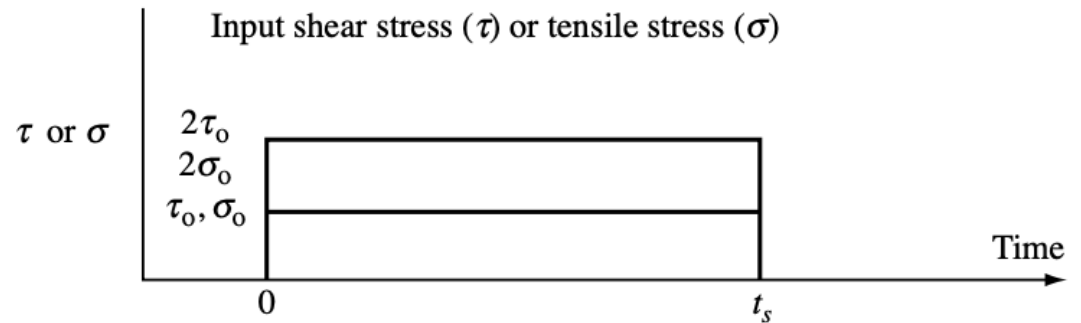
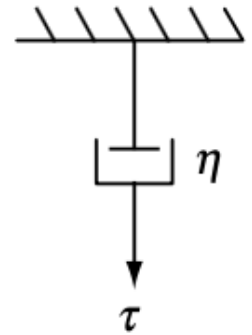
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- Constitutive equations for Maxwell, Kelvin/Voigt, Burgers models?

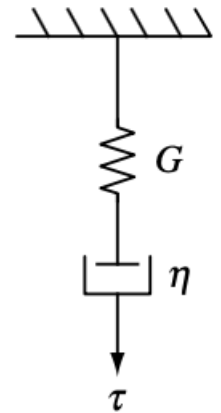
Spring model



Dashpot model

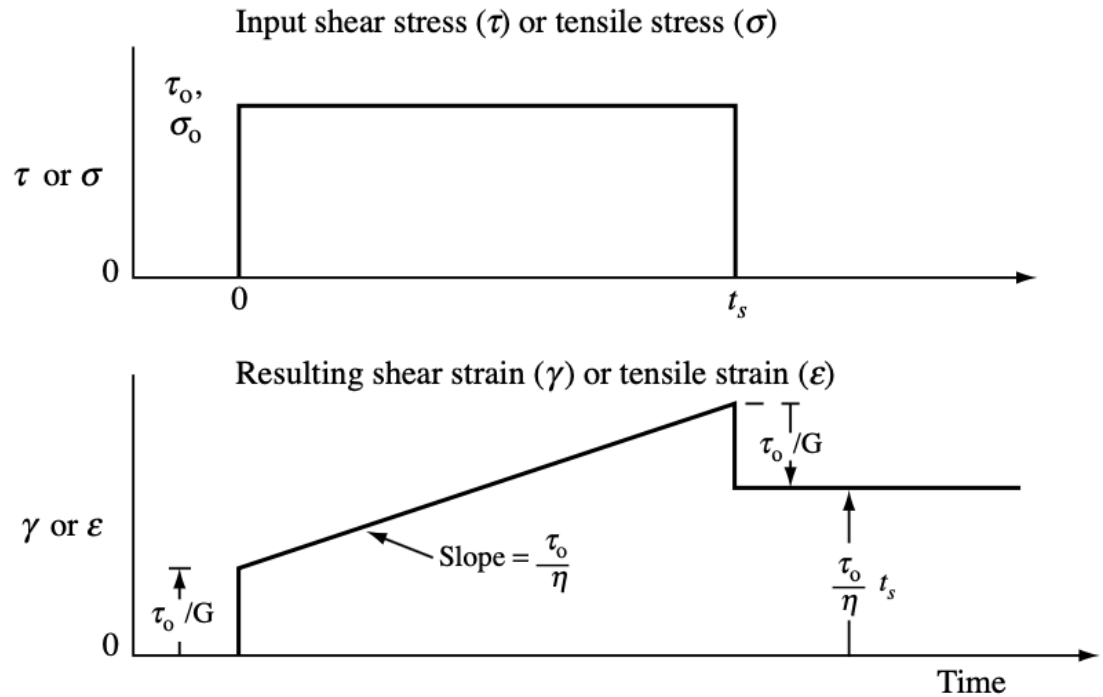


Maxwell model

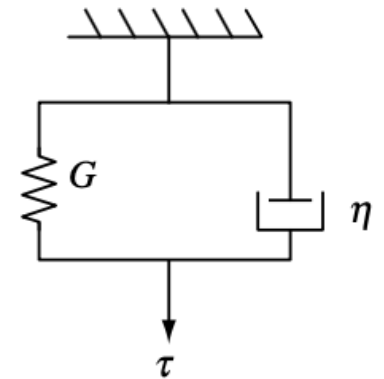


$$\varepsilon(t) = \sigma_o/E + \sigma_o/\eta_e t$$

$$\gamma(t) = \tau_o/G + \tau_o/\eta t$$

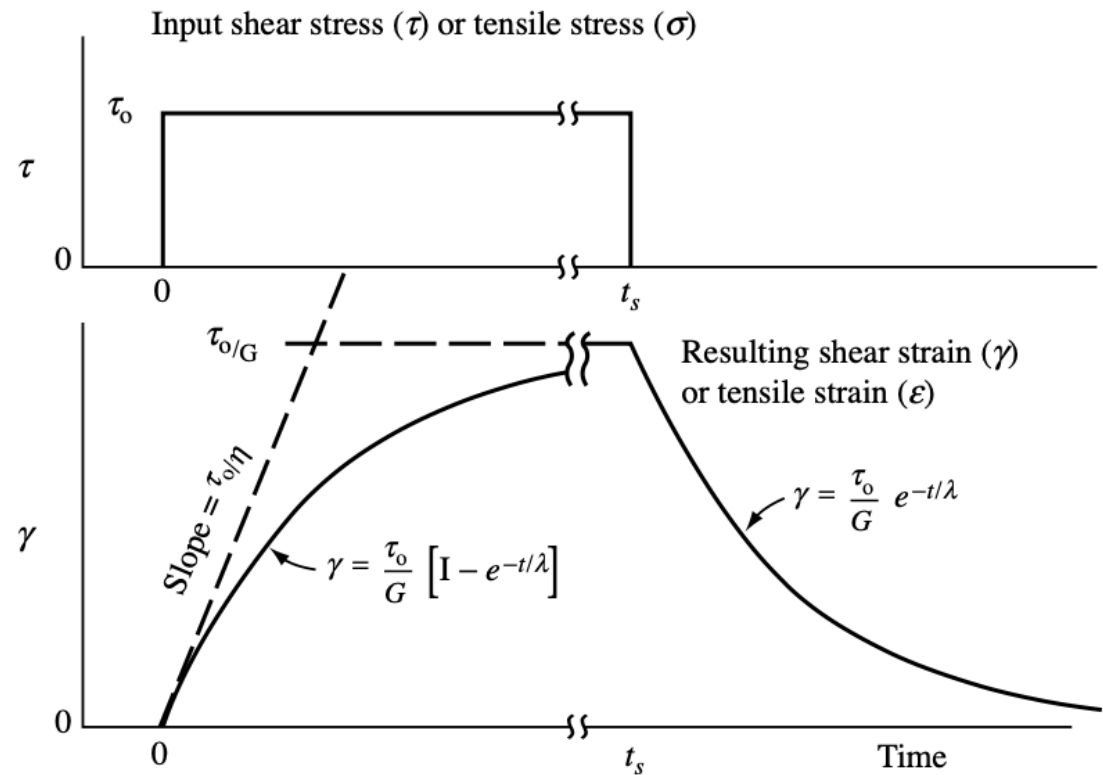


Kelvin/Voigt model

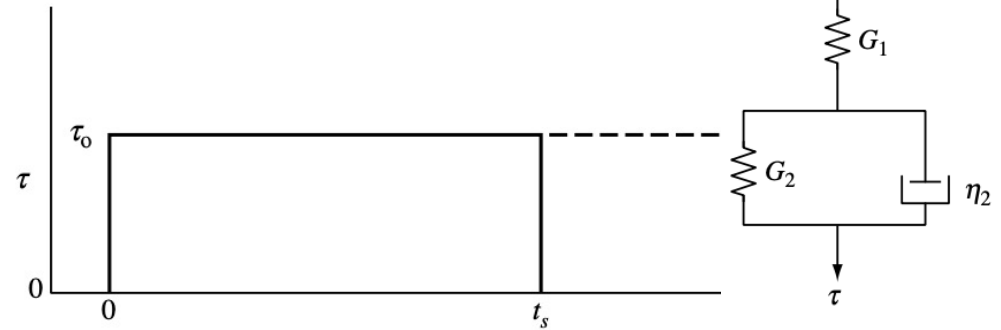


$$\gamma(t) = \frac{\tau_0}{G} (1 - e^{-t/\lambda})$$

$$J_c(t) = \frac{1}{G} (1 - e^{-t/\lambda})$$

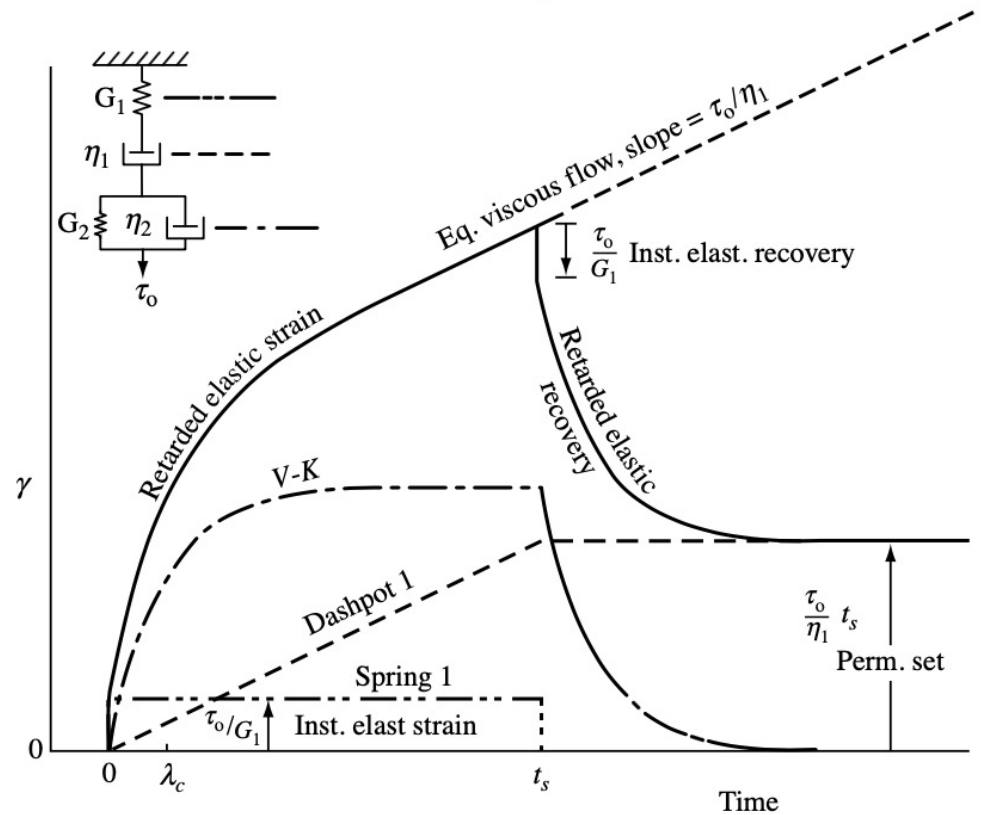


Burgers model



$$\gamma(t) = \frac{\tau_0}{G_1} + \frac{\tau_0}{\eta_1} t + \frac{\tau_0}{G_2} \left[1 - e^{-(G_2/\eta_2)t} \right]$$

$$J_e(t) = \frac{1}{G_1} + \frac{1}{\eta_1} t + \frac{1}{G_2} \left[1 - e^{-(G_2/\eta_2)t} \right]$$



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- Can different models be used together, how do they actually combine?

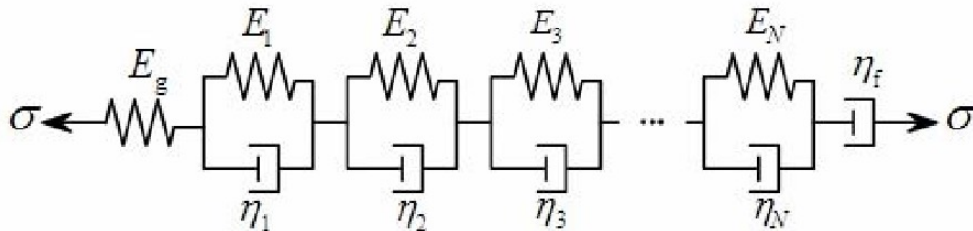


Figure 12. Generalized Voigt or Kelvin model

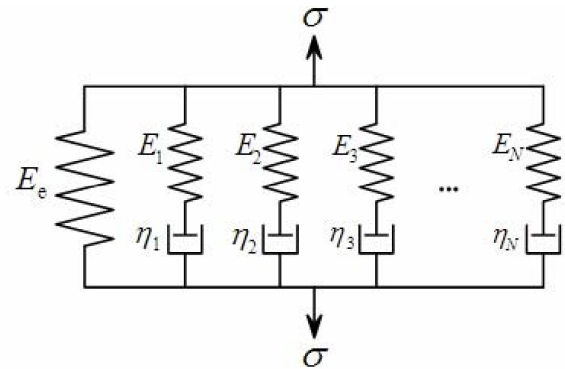
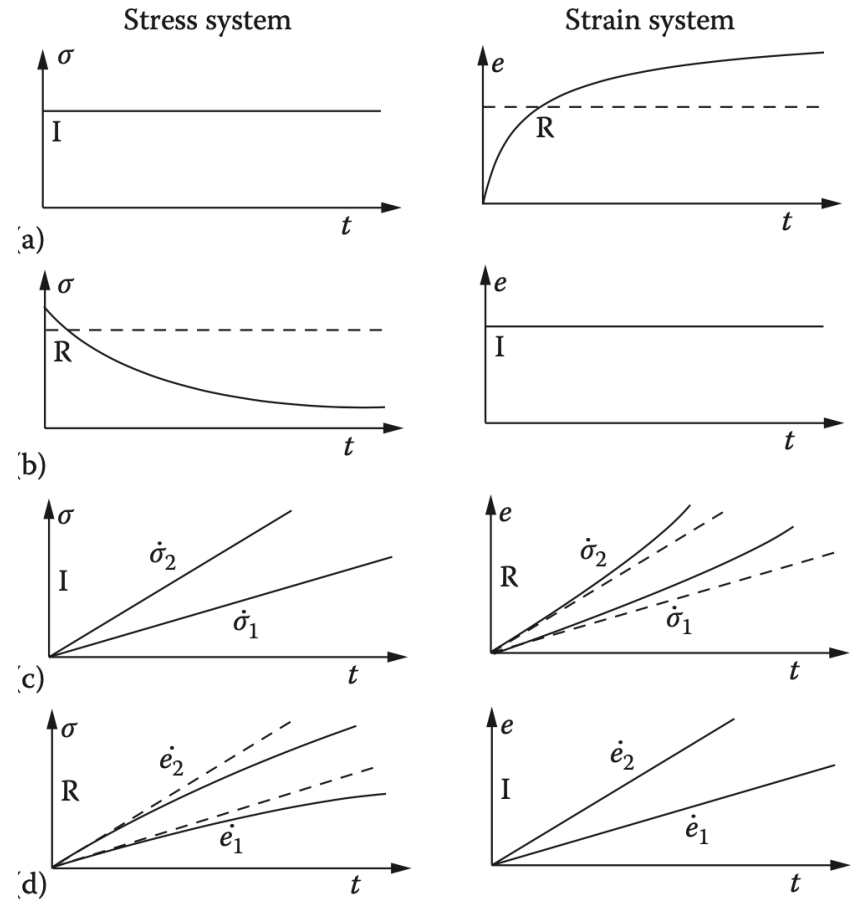


Figure 11. Generalized Maxwell or Wiechert model

Viscoelasticity

- If a constant stress, σ_1 or σ_2 , is applied to a viscoelastic material over time, t , strain, ε_1 evolves as shown in (a)
- Hence at an arbitrary time, t , the strains at the two stresses are $\varepsilon_1(t)$ and $\varepsilon_2(t)$, then;

$$\frac{\varepsilon_1(t)}{\sigma_1} = \frac{\varepsilon_2(t)}{\sigma_2}$$



Viscoelasticity

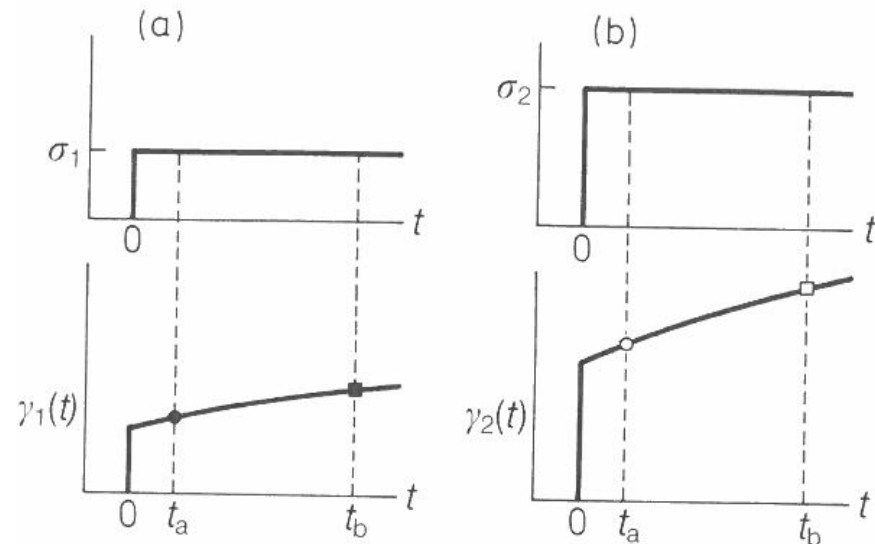
- If a constant stress, σ_1 or σ_2 , is applied to a viscoelastic material over time, t , strain, ε_1 evolves as shown in (a)
- Hence at an arbitrary time, t , the strains at the two stresses are $\varepsilon_1(t)$ and $\varepsilon_2(t)$, then;

$$\frac{\varepsilon_1(t)}{\sigma_1} = \frac{\varepsilon_2(t)}{\sigma_2}$$

- Strain in the two experiments are proportional to imposed stress. Hence creep compliance at time t can be defined as;

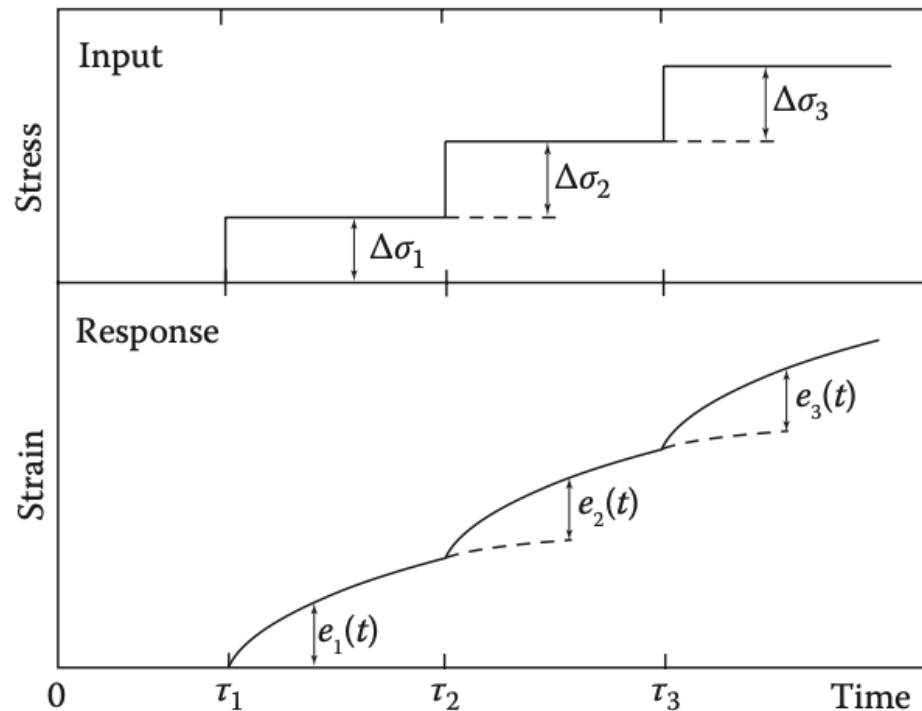
$$J(t) = \frac{\varepsilon_1(t)}{\sigma_1} = \frac{\varepsilon_2(t)}{\sigma_2}$$

in general, $J(t) = \frac{\varepsilon(t)}{\sigma}$



Boltzmann Superposition Principle

- Helps determine **state of stress or strain** in viscoelastic materials from knowledge of the **materials deformation history**
- For varying stress, overall deformation is related to the sum of strains due to each stress step.



Boltzmann Superposition Principle

- Helps determine **state of stress or strain** in viscoelastic materials from knowledge of the **materials deformation history**
- For varying stress, overall deformation is related to the sum of strains due to each stress step.

- We define creep compliance as

$$J(t) = \varepsilon(t)/\sigma, \quad \text{which is only a function of time}$$

then for each loading,

$$\varepsilon(t) = \Delta\sigma J(t - \tau)$$

- With further stressing we get;

$$\varepsilon(t) = \varepsilon_1(t) + \varepsilon_2(t) + \varepsilon_3(t) + \dots$$

$$= \Delta\sigma_1 J(t - \tau_1) + \Delta\sigma_2 J(t - \tau_2) + \dots$$

$$= \sum_{n=0}^n J(t - \tau) \Delta\sigma_n$$

Boltzmann Superposition Principle

- Helps determine **state of stress or strain** in viscoelastic materials from knowledge of the **materials deformation history**
- For varying stress, overall deformation is related to the sum of strains due to each stress step.
- *Provided there is a linear relationship between stress and strain and their time derivatives, then the strain produced by any set of applied stresses is the sum of the strains produced by the individuals of the set when acting alone.*

Boltzmann Superposition Principle

- We have seen that overall deformation (strain) is given by;

$$\varepsilon(t) = \sum_{n=0}^n J(t - \tau) \Delta \sigma_n$$

- Which can be represented as an integral ($-\infty$ to t = all deformation history);

$$\varepsilon(t) = \int_{-\infty}^t J(t - \tau) d\sigma(t)$$

- This equation can be express in terms of relaxation time, τ ;

$$\varepsilon(t) = \int_{-\infty}^t J(t - \tau) \frac{d\sigma(\tau)}{d\tau} d\tau$$

Predicts overall strain
After stressing the sample

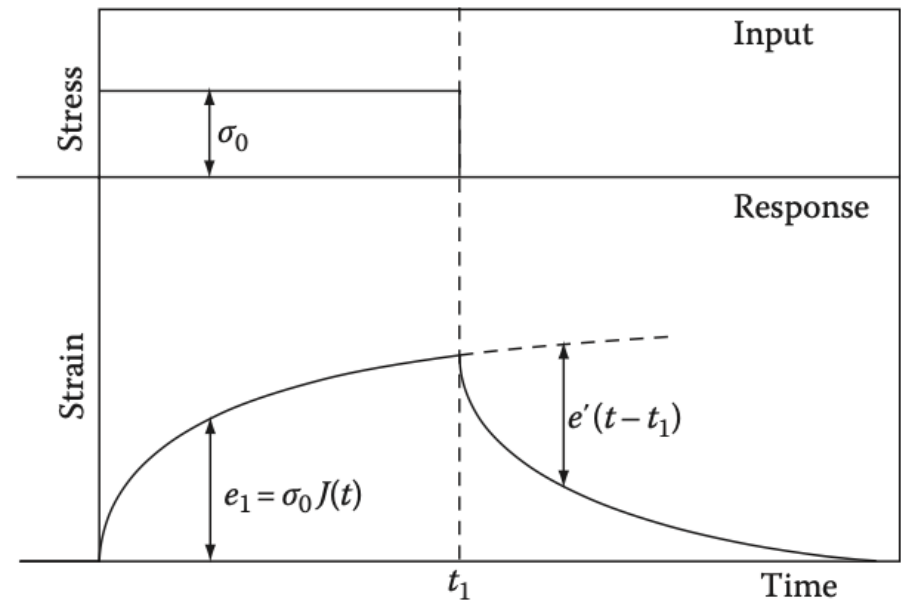
Boltzmann Superposition Principle: Recovery

$$e_1 = \sigma_0 J(t)$$

$$e_2 = -\sigma_0 J(t - t_1)$$

$$e(t) = \sigma_0 J(t) - \sigma_0 J(t - t_1)$$

$$e'(t - t_1) = \sigma_0 J(t - t_1)$$



Boltzmann Superposition Principle

- We can also consider stress relaxation the same way:

$$G(t) = \frac{\sigma(t)}{\varepsilon}$$

- If we consider the stress relaxation modulus, then;

$$\sigma(t) = G(t)\varepsilon = \Delta\varepsilon_1 G(t - \tau_1) + \Delta\varepsilon_2 G(t - \tau_2) + \dots$$

- This is analogous to what we saw above, hence;

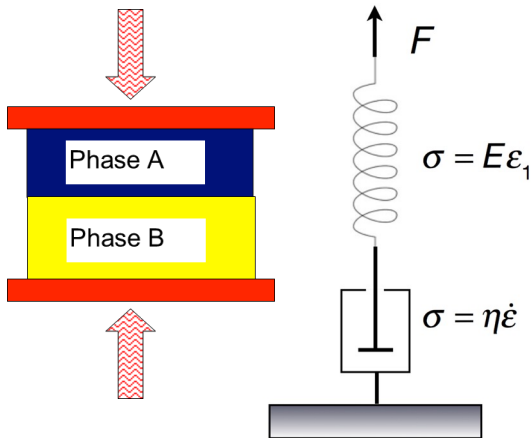
$$\sigma(t) = \int_{-\infty}^t G(t - \tau) \frac{d\varepsilon(\tau)}{d\tau} d\tau$$

Predicts overall stress
After straining the sample

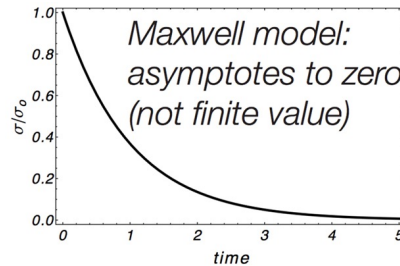
Recall:

Elasticity (chpt. 19)

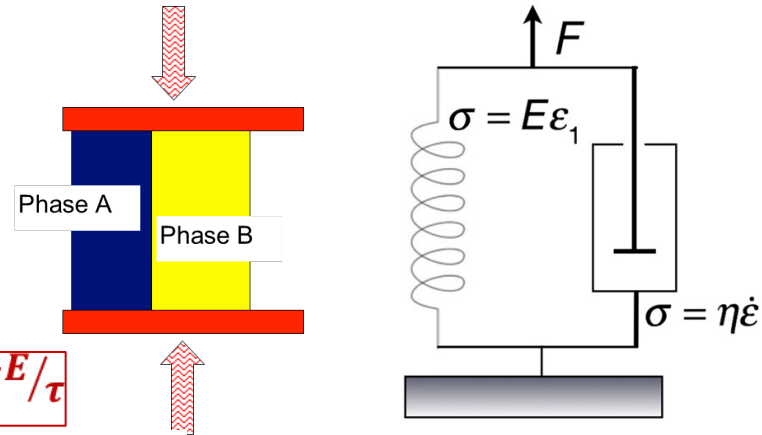
Maxwell-Reuss (isostress)



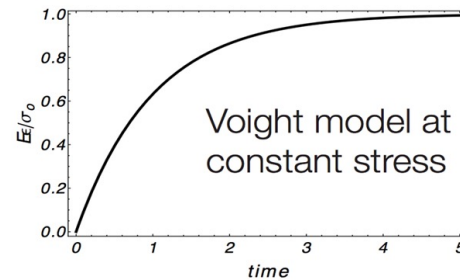
$$\sigma = \sigma_0 e^{-(E/\eta)t}$$



Kelvin-Voigt (isostrain)



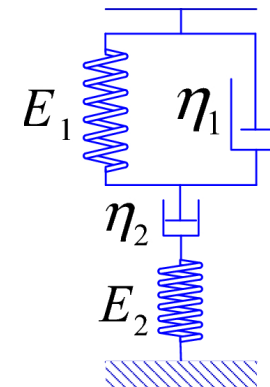
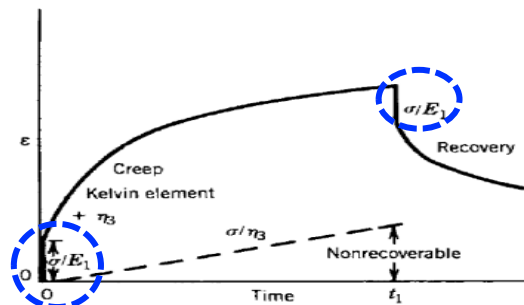
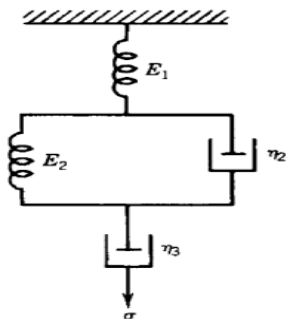
$$\epsilon = \epsilon_0 e^{-E/\tau}$$



Other Model

- Three-element (zener) model – overall strain can be expressed as;

$$\epsilon = \frac{\sigma}{E_1} + \frac{\sigma}{E_2} \left(1 - e^{-t/\tau_2}\right) + \frac{\sigma}{\eta_3} t$$



(d) Burgers Model

Revisiting Creep Compliance

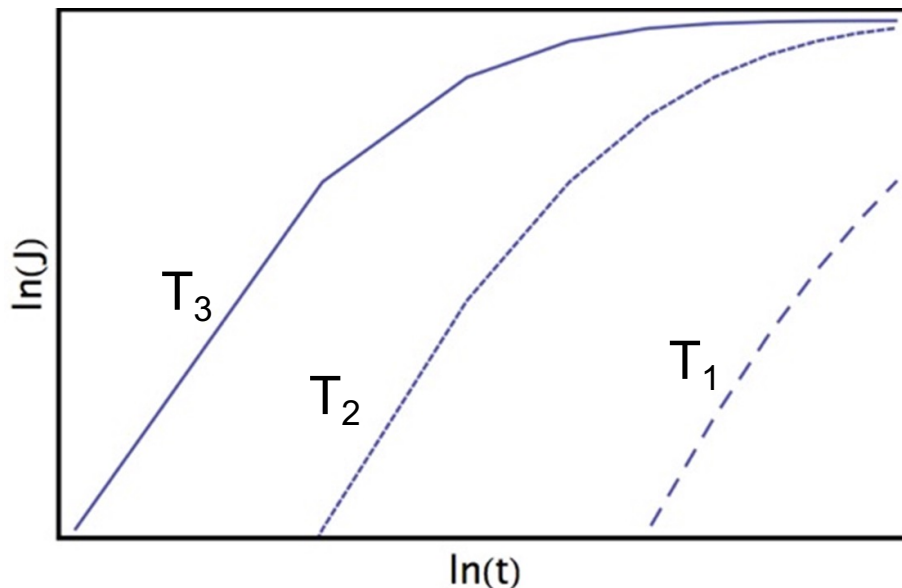
- A better way to represent creep is to normalize strain with a constant applied load (i.e. creep compliance);

$$J(t) = \frac{\varepsilon(t)}{\sigma_0}$$

- But from Kelvin/Voigt model;

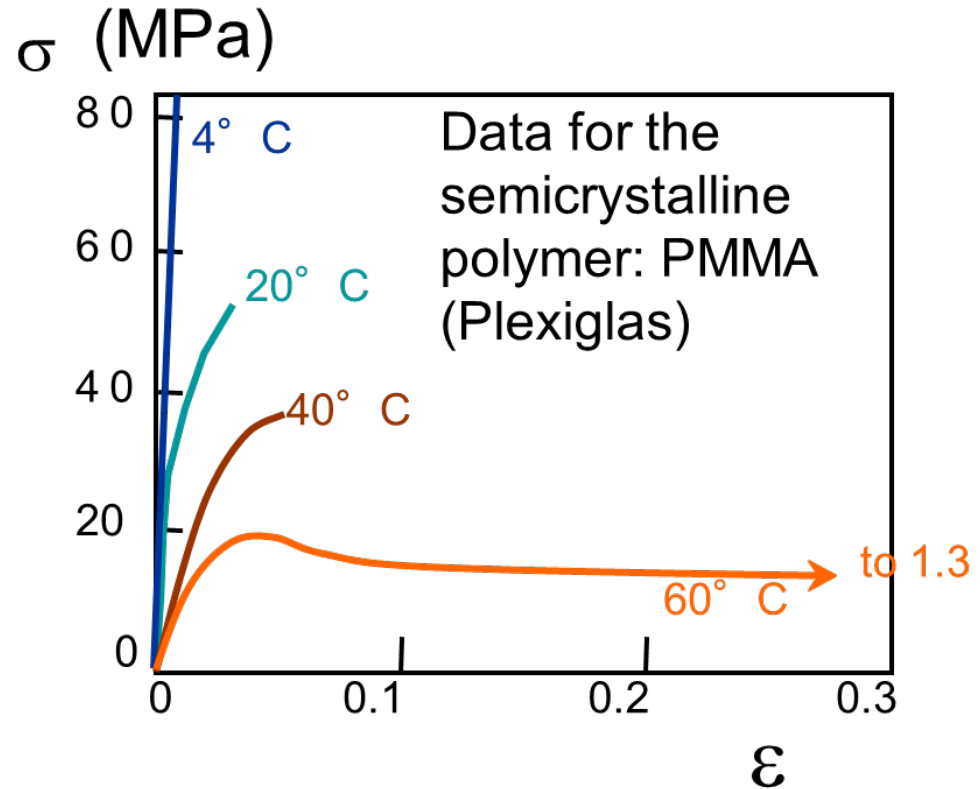
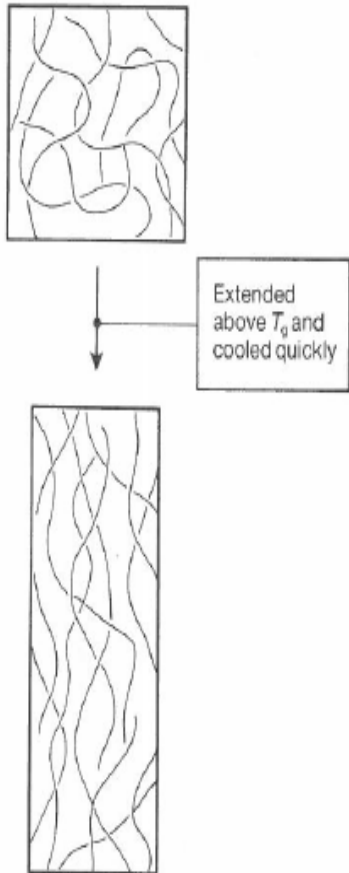
$$\varepsilon = \left(\frac{\sigma_0}{E}\right) \left[1 - e^{-t/\tau}\right] \Rightarrow \boxed{\frac{\varepsilon(t)}{\sigma_0}} = \frac{1}{E} \left(1 - e^{-t/\tau}\right)$$

- A better way to represent this is in log form, hence plot $\log(J)$ vs $\ln(t)$



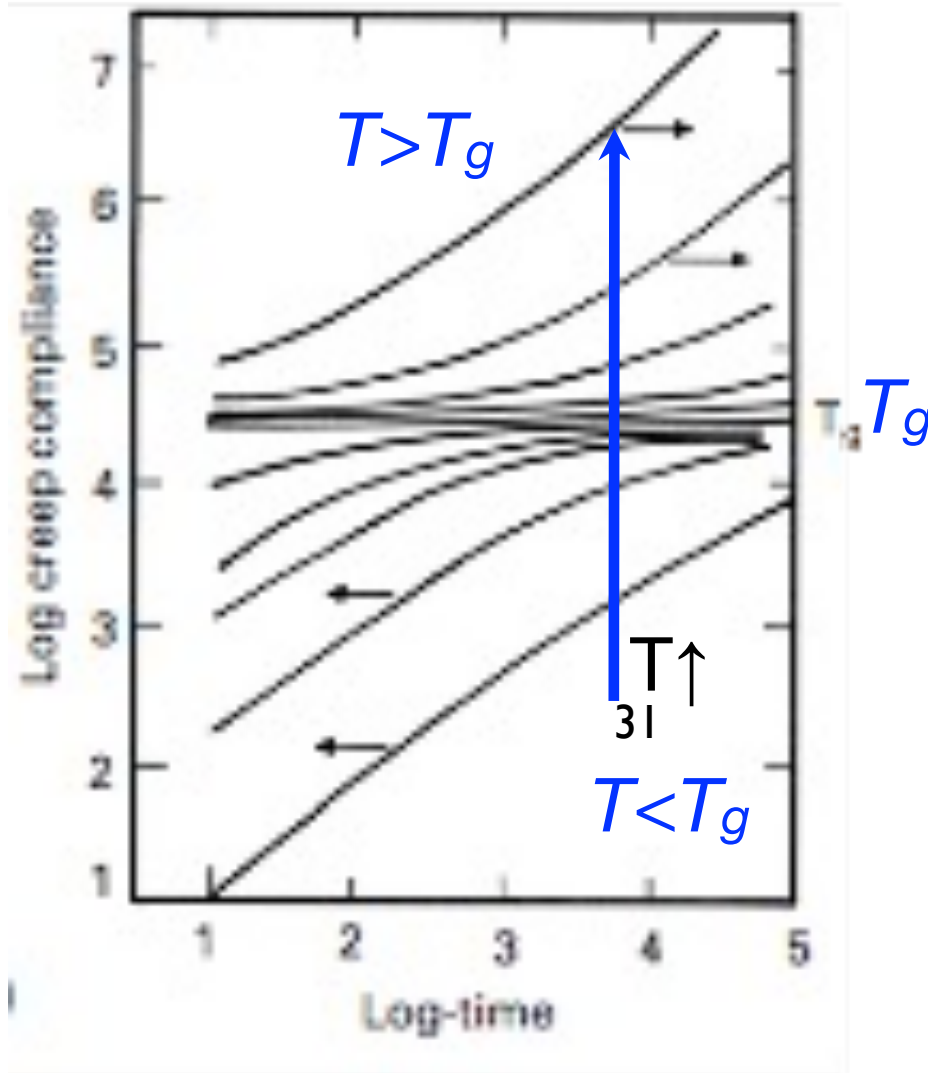
- Since relaxation time is dependent on temperature, we see different curves at different T .

...But, properties depends on temperature too



- What is the basis for this thermal-driven change in mechanical properties?
- How do we predict thermal response of a material under stress?
- First look at T -driven *change in stress at constant elongation*.

Creep compliance scaling



Creep compliance curves can be scaled onto a single “master curve” by

- shifting each curve by an amount that depends on T
- the $\ln(t)$ values shifted by a temperature dependent parameter of the form $\log(a_t)$: now plot vs. $\log(t/a_t)$
- for $T < T_g$, curves shifted to left
- for $T > T_g$, curves shifted to right

Time-Temperature Superposition

Williams-Landel-Ferry (WLF) Equation (time-temperature superposition)

- Its possible to interrelate time & temperature dependence of viscoelasticity
- Flow, a molecular relaxation effect, requires free volume, but in polymers this is limited by neighboring polymer chains
- Doolittle's theory of viscosity relates viscosity to occupied volume and free volume;

$$\ln \eta = B \left(\frac{v_o - v_f}{v_f} \right) + \ln A$$

A & B are constants

$$\frac{v_f}{v_o} \simeq \frac{v_f}{v_o + v_f} = f$$

- The WLF compares time dependent behavior of a polymer at two different temperatures

$$\ln a_T = - \frac{\left(B / f_o \right) (T - T_o)}{f_o / \alpha_f + (T - T_o)}$$

α_f = expansion coefficient of the free volume

f_o = fractional free volume at T_g or other point of interest

- But (B / f_o) and f_o / α_f are constants, therefore;

$$\ln a_T = - \frac{c_1 (T - T_o)}{c_2 + (T - T_o)}$$

T_o is the reference temperature

$$\frac{v_f}{v_o} \simeq \frac{v_f}{v_o + v_f} = f$$

Williams-Landel-Ferry (WLF) equation

empirically:

$$\log(a_t) = -c_1 \frac{T - T_s}{c_2 + T - T_s}$$

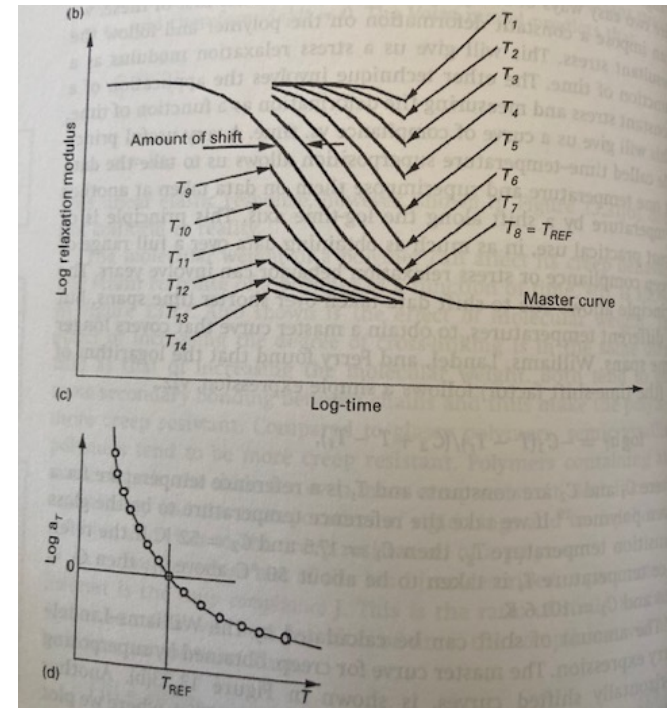
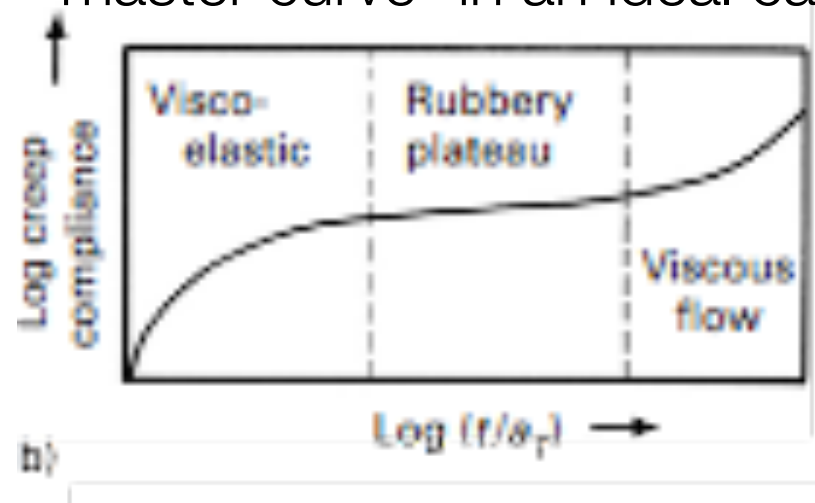
$$\log(a_t) = -c_1 \frac{T - T_s}{c_2 + T - T_s}$$

T_s is a reference temperature and c_1 and c_2 are constants.

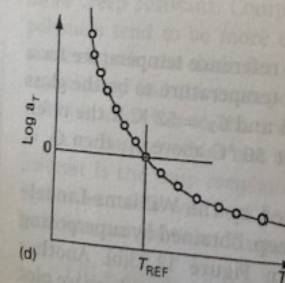
With $T_s = T_g$, $c_1 = 17.5$ and $c_2 = 52$ C.

t is the time it takes to reach a given stress level

“master curve” in an ideal case



(c) Log-time



(d)

Constructing a master curve

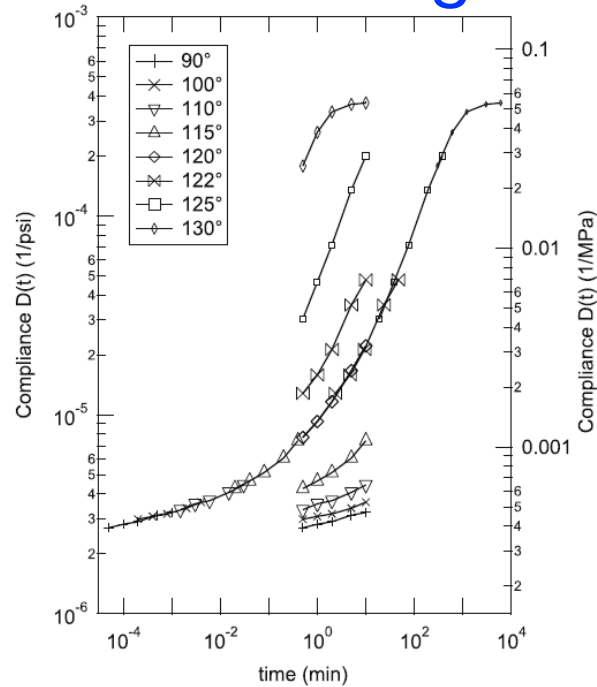
Hal F. Brinson · L. Catherine Brinson

Polymer
Engineering
Science and
Viscoelasticity

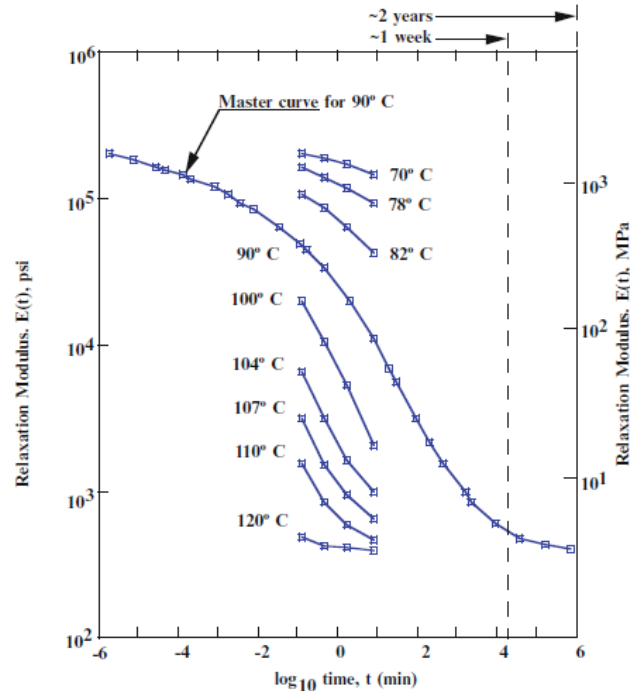
An Introduction

Second Edition

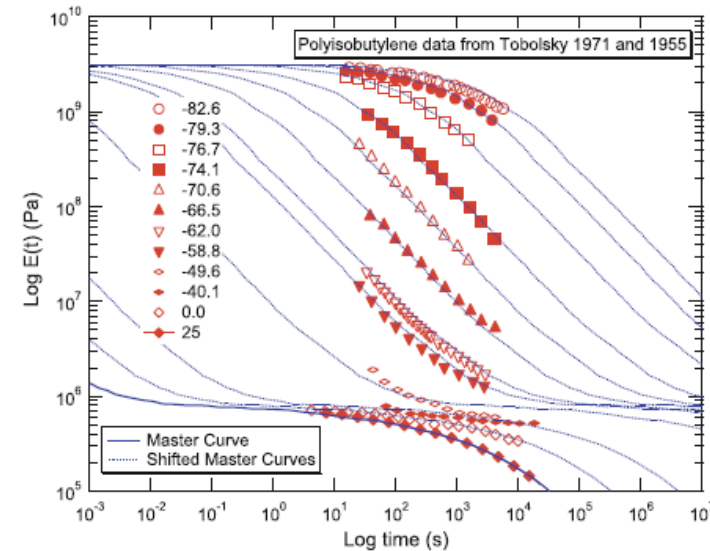
Springer



modified (rubber toughened) epoxy adhesive

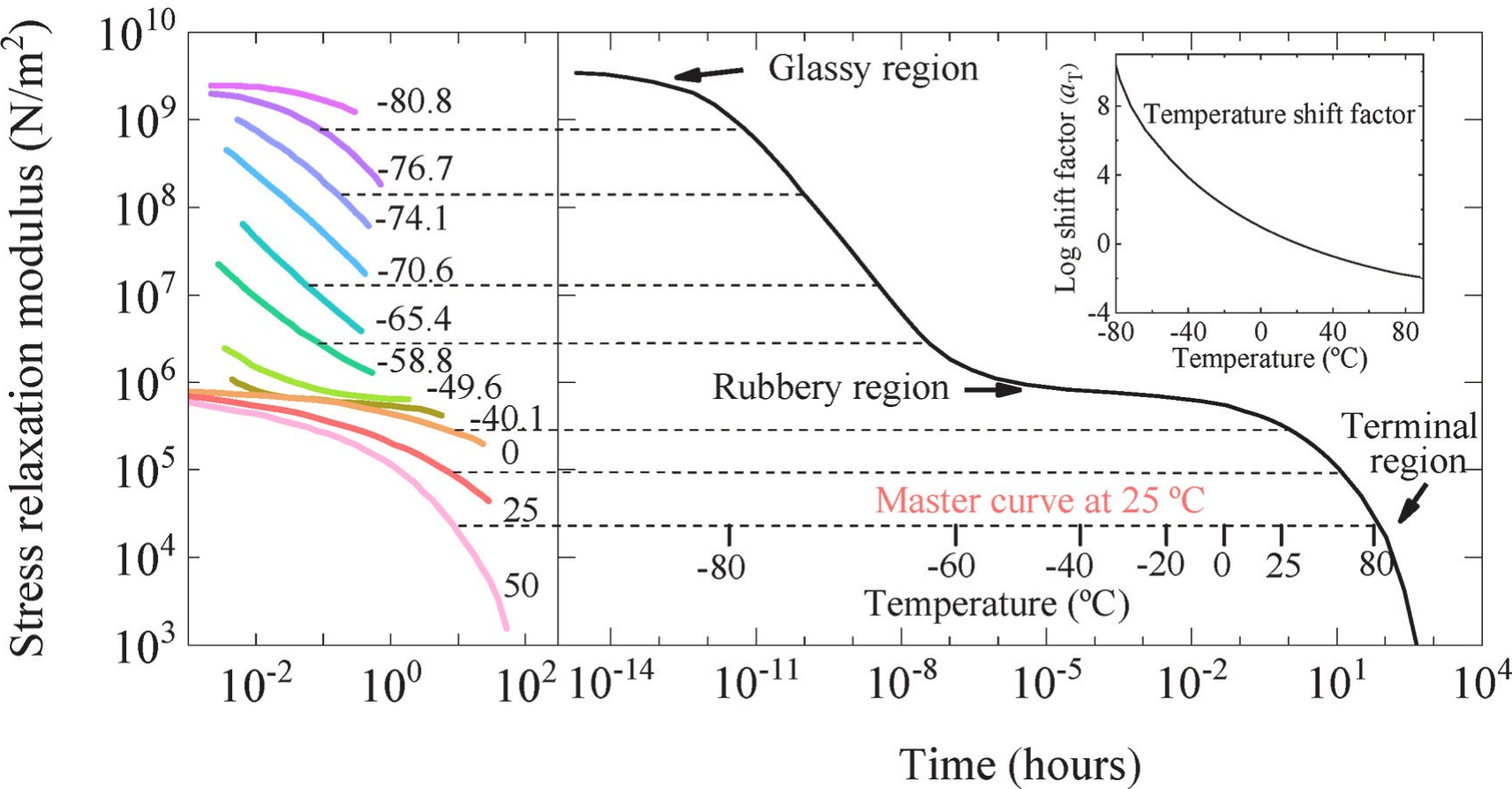


Polyisobutylene



Creep compliance (D instead of J here)

Constructing a master curve



Example 2

Suppose we have data on a stress relaxation modulus vs. time for an amorphous polymer (like chewing gum). Suppose that the polymer has a glass transition temperature of $T_g=100\text{ }^\circ\text{C}$. We measure a creep modulus of 150 GPa of 1 hour at $95\text{ }^\circ\text{C}$. How much time would it take to achieve the same creep modulus at $90\text{ }^\circ\text{C}$?

With $T_s=T_g$, $c_1 = 17.5$ and $c_2 = 52\text{ }^\circ\text{C}$.

NOTE temperature measured in $^\circ\text{C}$

$$\begin{aligned}\log(a_t(T_2)) - \log(a_t(T_1)) &= \log\left(\frac{a_t(T_2)}{a_t(T_1)}\right) \\ &= \log\left(\frac{t(T_2)}{t(T_s)}\right) - \log\left(\frac{t(T_1)}{t(T_s)}\right) = \log\left(\frac{t(T_2)t(T_s)}{t(T_s)t(T_1)}\right) = \log\left(\frac{t(T_2)}{t(T_1)}\right)\end{aligned}$$

$$\log(t(T_2)) = \log(t(T_1)) + \log\left(\frac{a_t(T_2)}{a_t(T_1)}\right)$$

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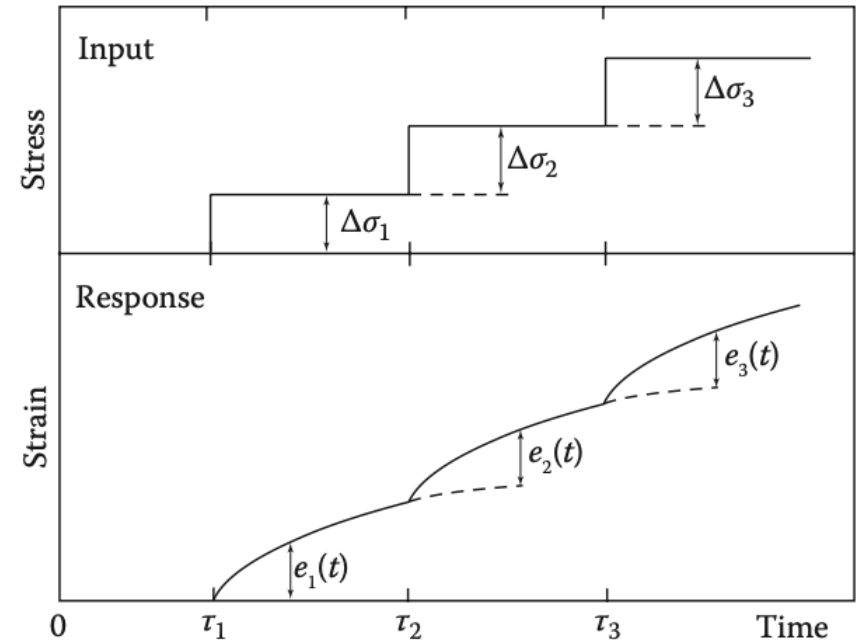
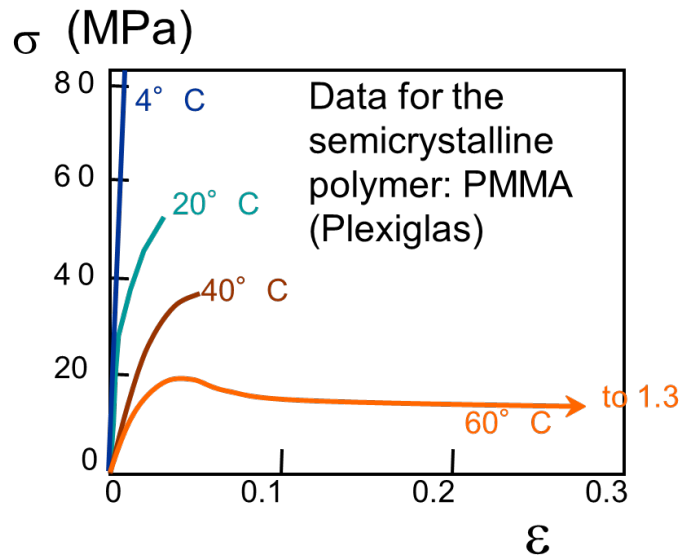
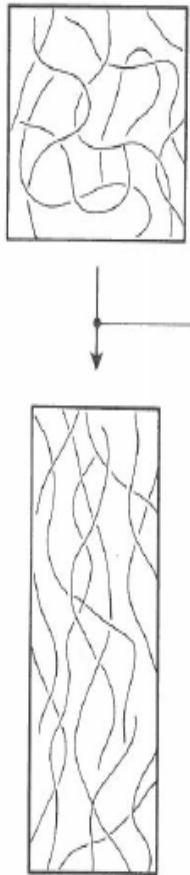
$$\log(t(T_2)) = \log(t(T_1)) + \log\left(\frac{a_t(T_2)}{a_t(T_1)}\right)$$

For our problem:

$$\log(a_{95}) = 1.86 \text{ and } \log(a_{90}) = 4.17 \Rightarrow \log\left(\frac{a_{90}}{a_{95}}\right) = 2.31$$

$$\log\left(\frac{t(90)}{1 \text{ hour}}\right) = 2.31 \Rightarrow t(90) = 10^{2.24} \text{ hour} = 204 \text{ hour}$$

...But, properties depends on temperature too



Boltzmann's principle of superposition states that, provided there is a linear relationship between stress and strain and their time derivatives, then the strain produced by any set of applied stresses is the sum of the strains produced by the individuals of the set when acting alone.